ME 282

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals

- A four-axis coordinate system otherwise called Miller-Bravais system is employed
- Directional indices are denoted by 4 indices, as ***[u v t w]***
- By convention, the first 3 indices pertain to the projections along the a_1, a_2, a_3 axes in the basal plane
- Conversion from a 3 index system to a 4 index system is achieved using the following formulae

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals

- $[\,u'\,v'\,w'] \rightarrow [\,u\,v\,t\,w]$

$$u = \frac{1}{3}(2u' - v') \qquad v = \frac{1}{3}(2v' - u')$$

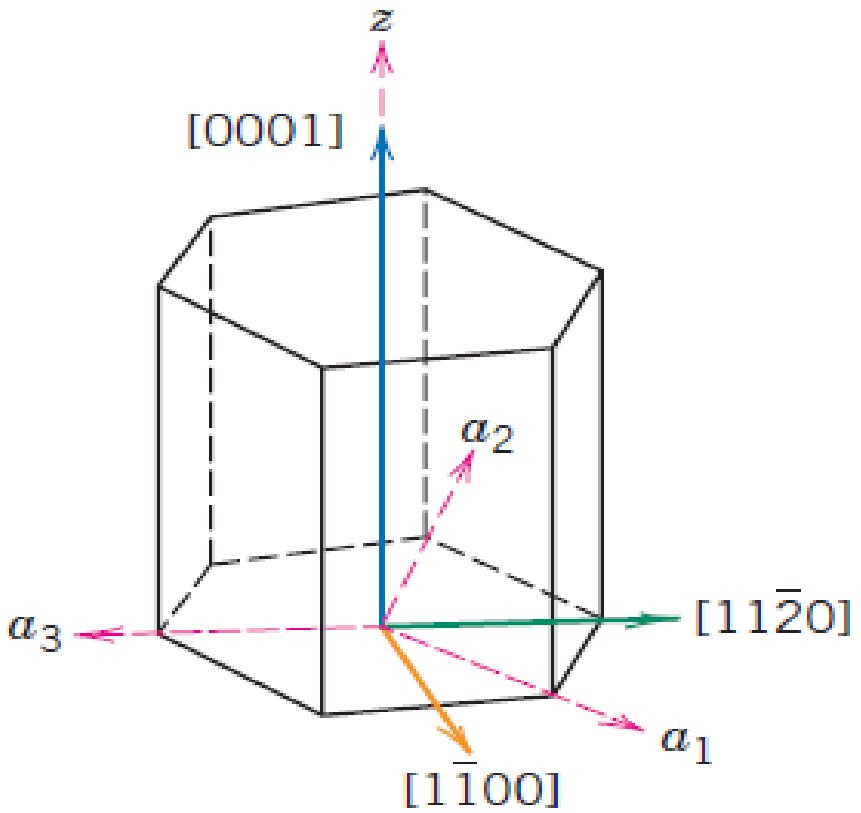
$$t = -(v + u) \qquad w = w'$$

- Primed indices are associated with 3 index scheme
- Unprimed indices are associated with the Miller Bravais
- Four-index system [0 1 0] direction becomes $[\bar{1}\,2\,\bar{1}\,0]$

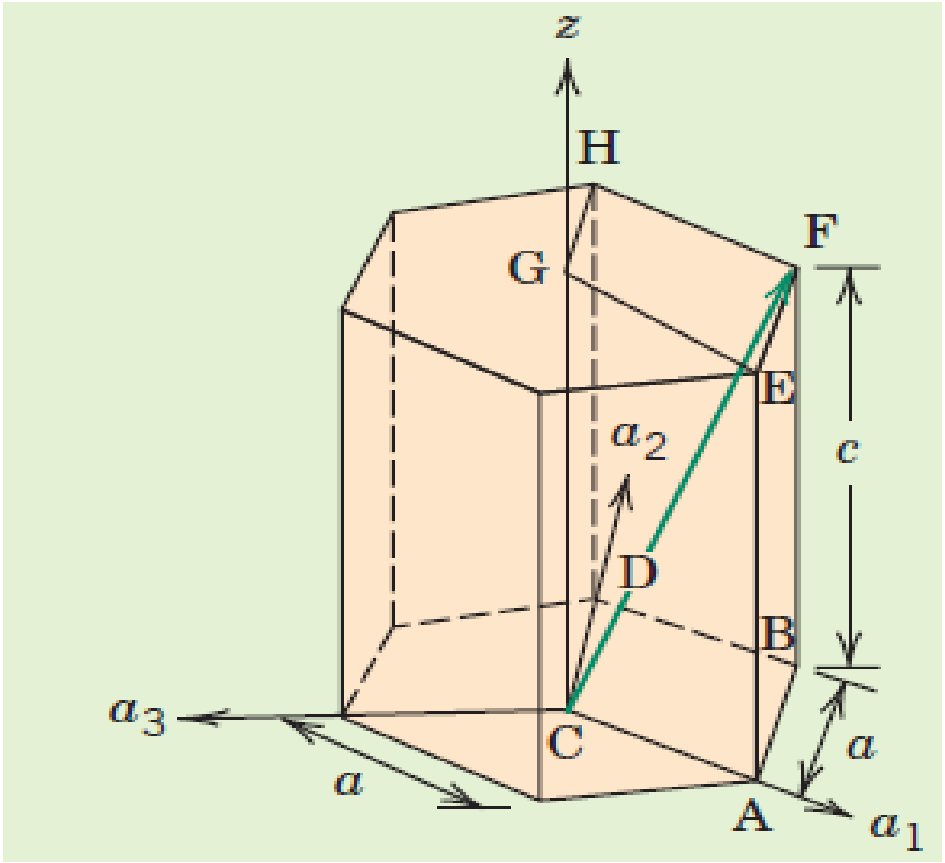
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals



(a)



(b)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Linear Density – For Directions

- Linear density (LD) is defined as the number of atoms per unit length whose centers lie on the direction vector for a specific crystallographic direction, i.e.

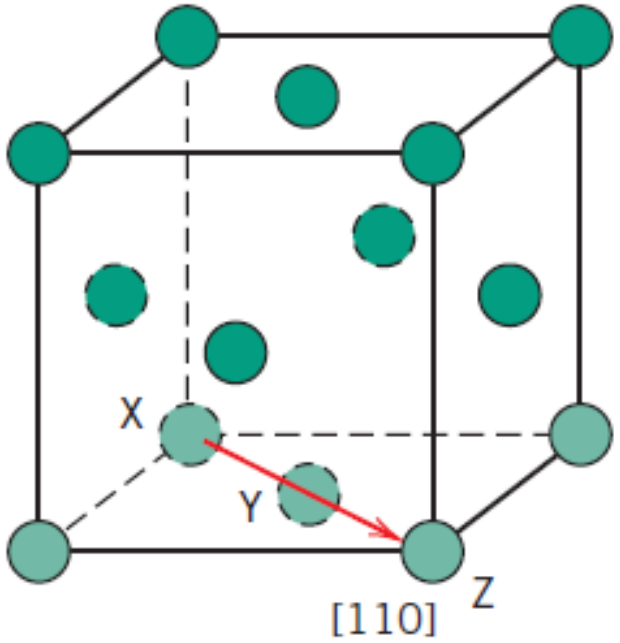
$$LD = \frac{number\ of\ atoms\ centered\ on\ direction\ vector}{length\ of\ direction\ vector}$$

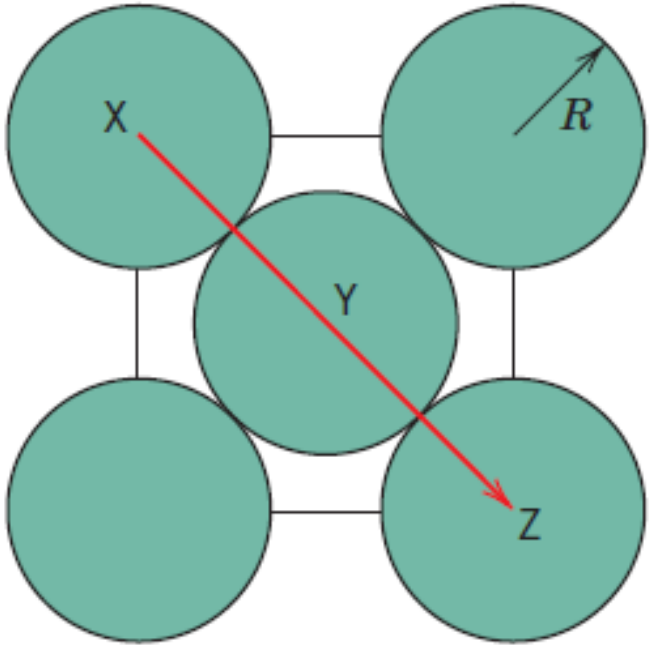
- Units (e.g. nm^{-1}, m^{-1})

THE STRUCTURE OF CRYSTALLINE SOLIDS

Linear Density

- Find the linear density for [1 1 0] within an FCC unit cell





$$LD_{110} = \frac{2 \text{ atoms}}{4R} = \frac{1}{2R}$$

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Planes

- The procedure used to obtain the *h*, *k*, *l* index numbers are as follows:

1. If the plane passes through the selected origin, either another parallel plane is constructed within the unit cell by an appropriate translation, or another origin is established at the corner of another unit cell

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Planes

2. At this point the plane either intersects or parallels each of the *3* axes; the length of the planar intercept for each of the axes is determined in terms of the lattice parameters *a*, *b* and *c*

3. The reciprocals of these numbers are taken. A plane that parallels an axis may be considered to have an infinite intercept, and, therefore has a zero index

4. If necessary, the 3 numbers are changed to the set of smallest integers by the multiplication or division by a common factor

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Planes

5. Finally, the integer indices, not separated by commas, are enclosed within parentheses, thus: *(h k l)*

- Intercepts on the negative axis has a bar over the particular index
- Reversing the directions of all indices specifies a plane parallel to, on the opposite side of and equidistance from, the origin
- For cubic crystals, planes and directions with the same indices are perpendicular to one another (who can show this using [1 0 0] and (1 0 0) ???)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Planes

- A family of planes contains all planes that are crystallographically equivalent, i.e. have the same atomic packing and is designated by indices enclose in braces, e.g., {100}

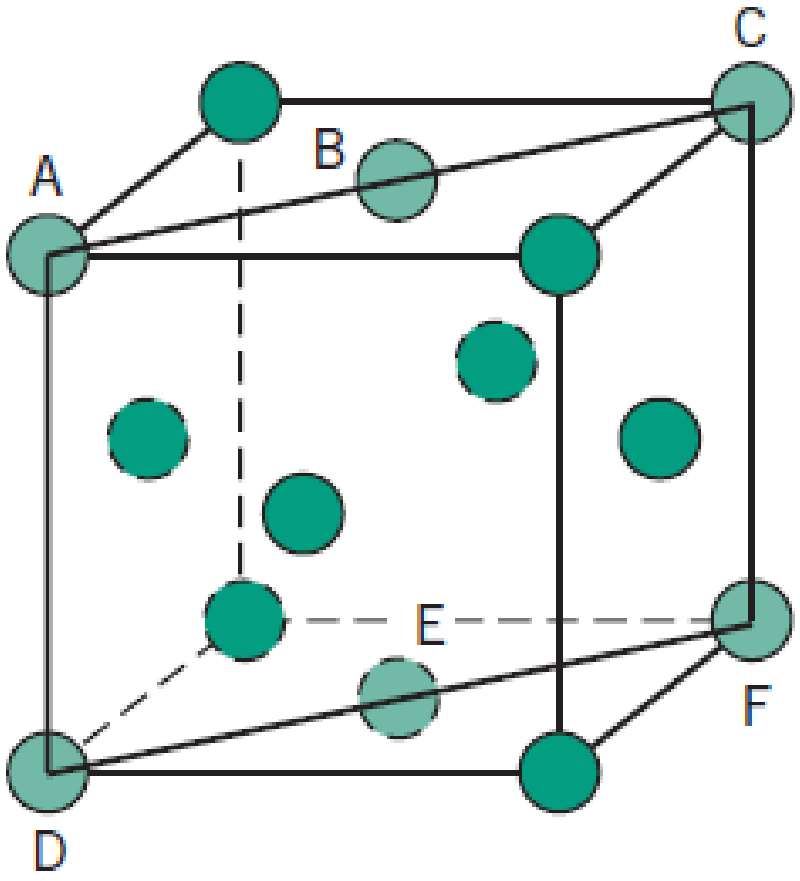
❖ *Atomic Arrangements for Planes*

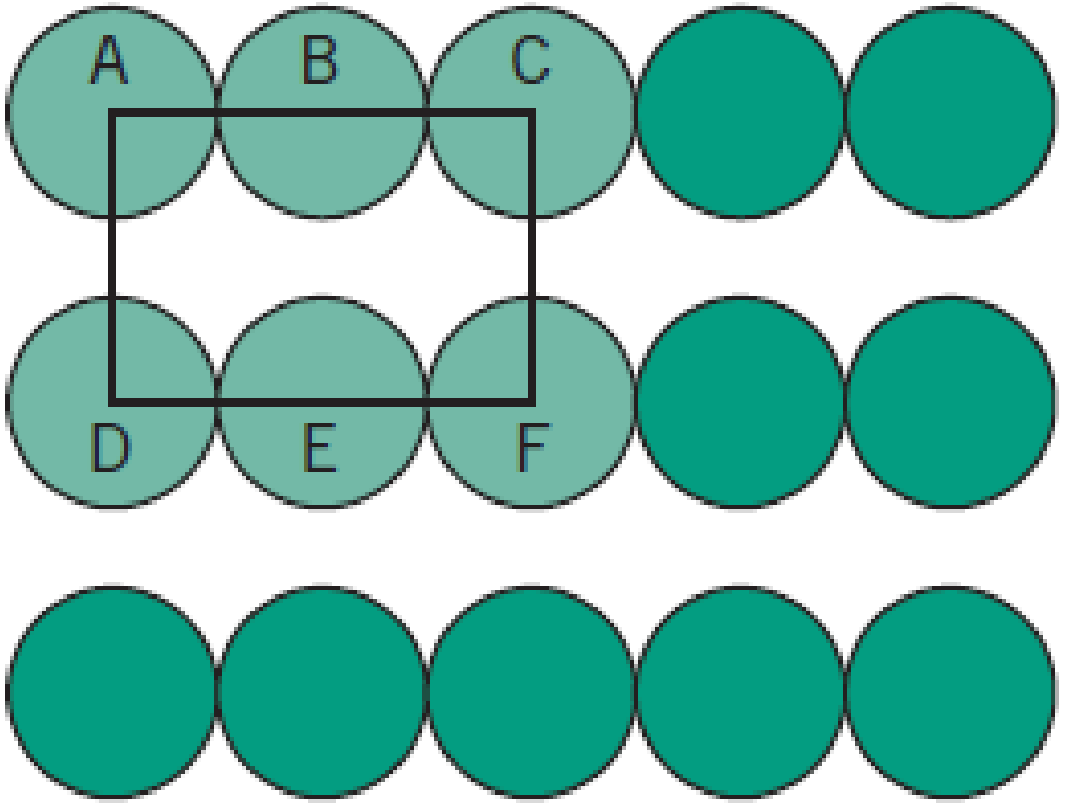
- Atomic arrangement for a crystallographic plane depends on the **type of crystal structure**
- Atoms here are represented by **circles** as would be obtained by a slice through the center of a full-sized hard sphere

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

Atomic Arrangements for (1 1 0) – FCC

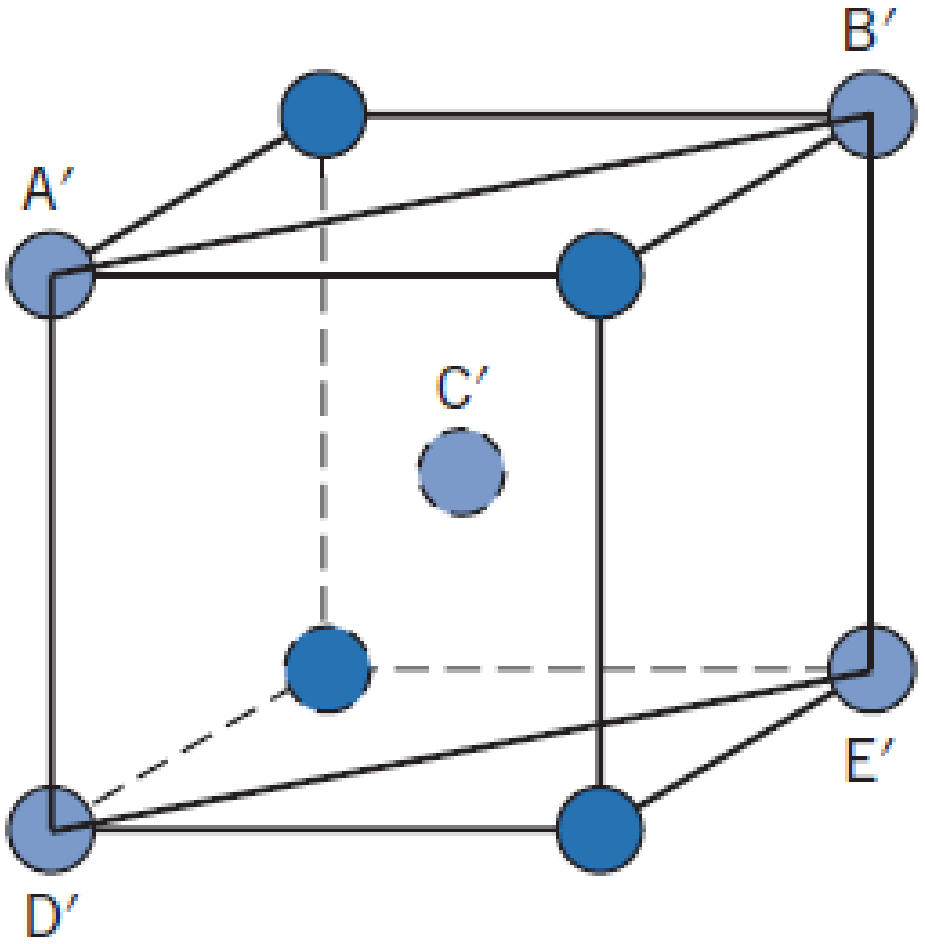


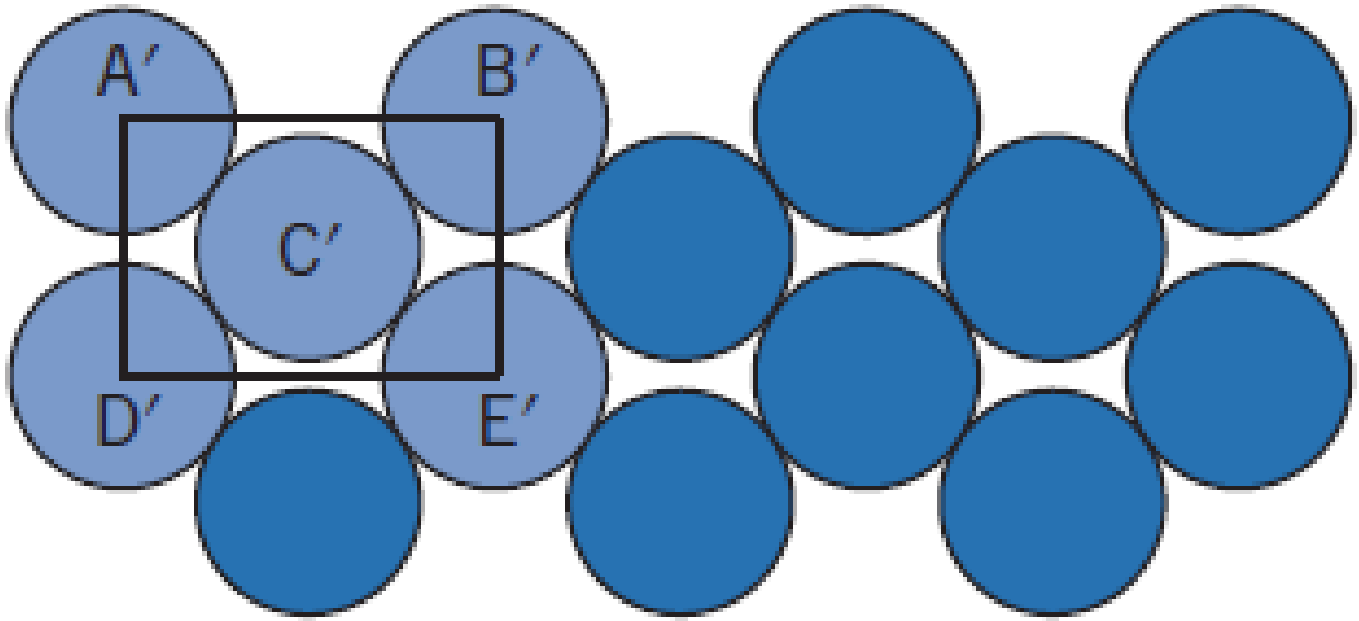


THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

Atomic Arrangements for (1 1 0) – BCC

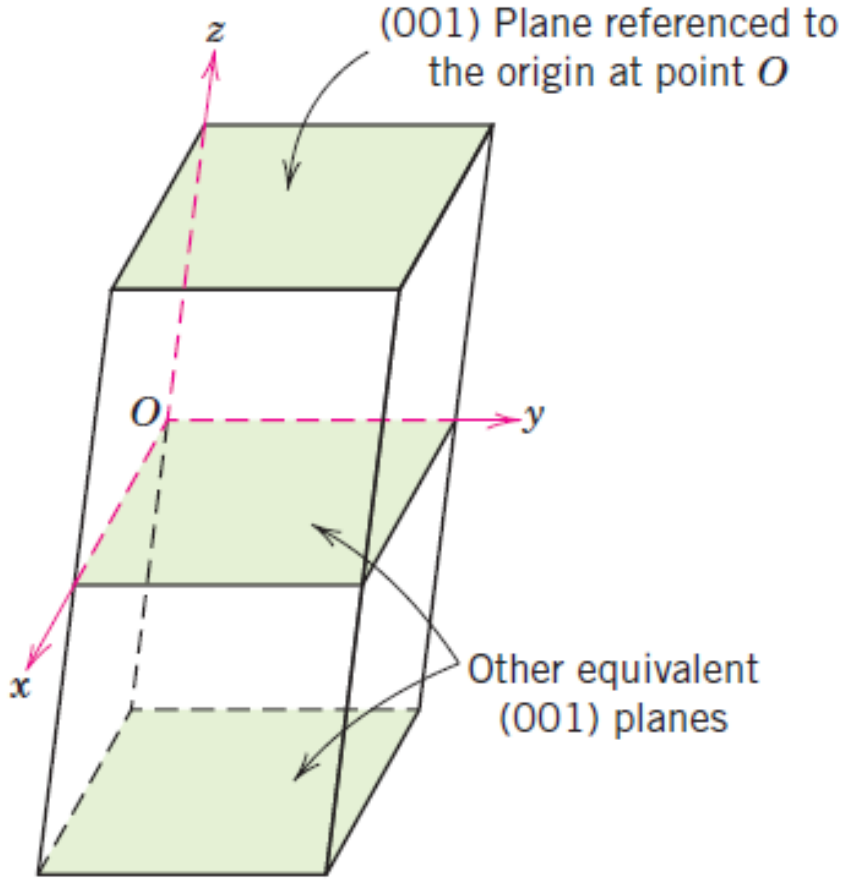


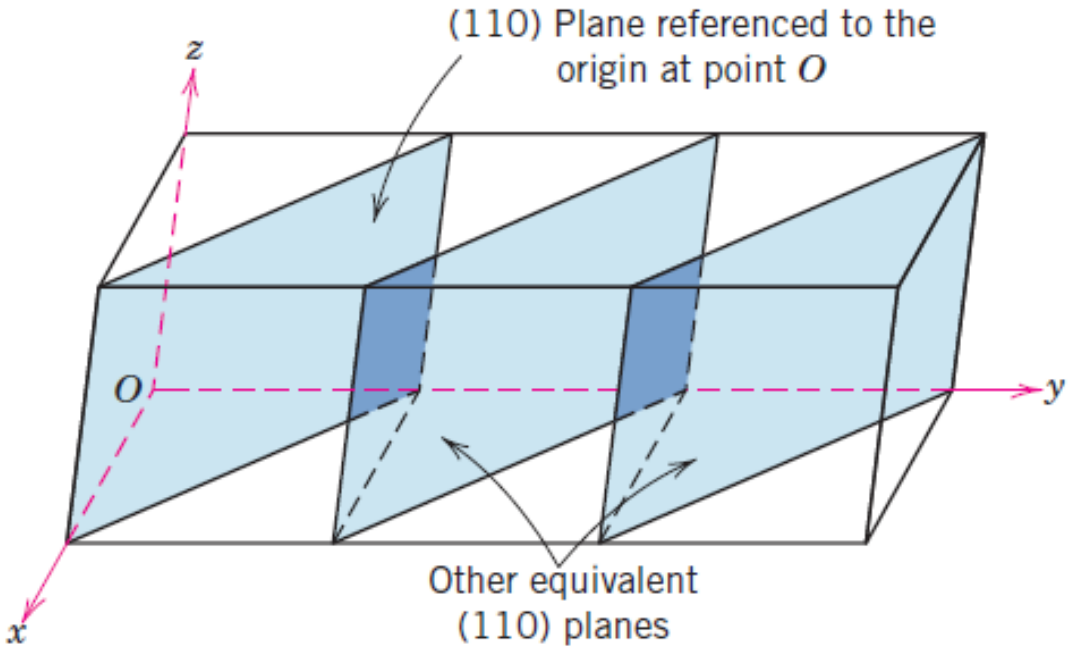


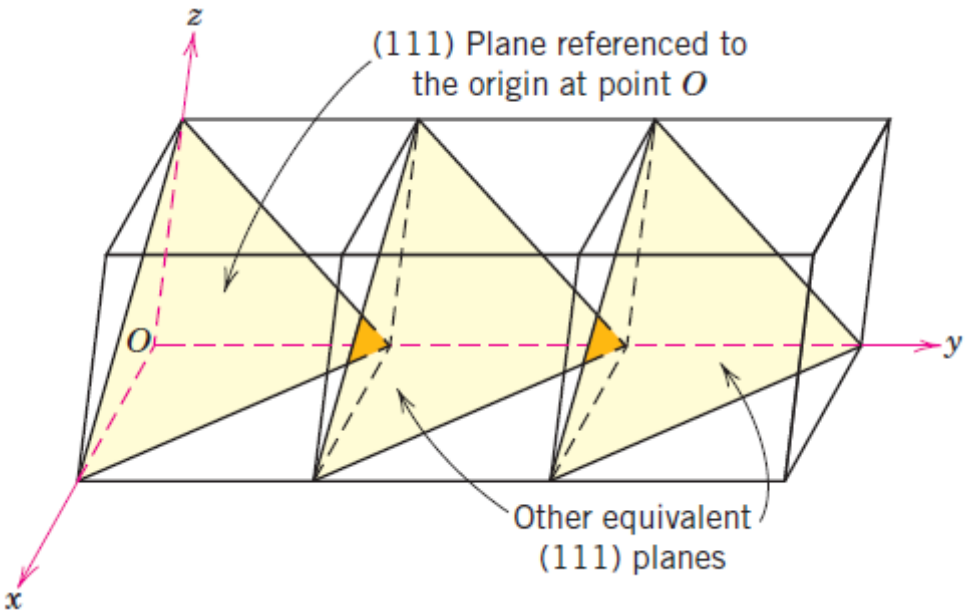
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

Some Equivalent Planes







THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

- Example, in cubic crystals, the planes (1 1 1), ($\bar{1}$ 1 1), (1 1 $\bar{1}$), (1 $\bar{1}$ 1), ($\bar{1}$ 1 $\bar{1}$), ($\bar{1}$ $\bar{1}$ 1), (1 $\bar{1}$ $\bar{1}$), and ($\bar{1}$ $\bar{1}$ $\bar{1}$) all belong to the same family {1 1 1}

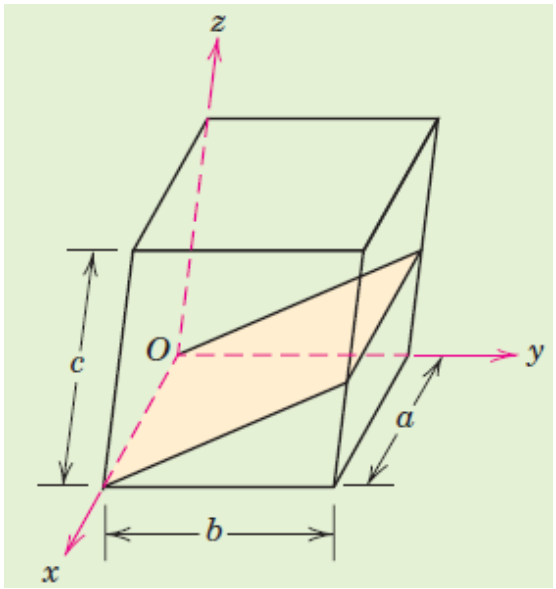
- Again in cubic systems, planes having the same indices, irrespective of the order or sign are equivalent

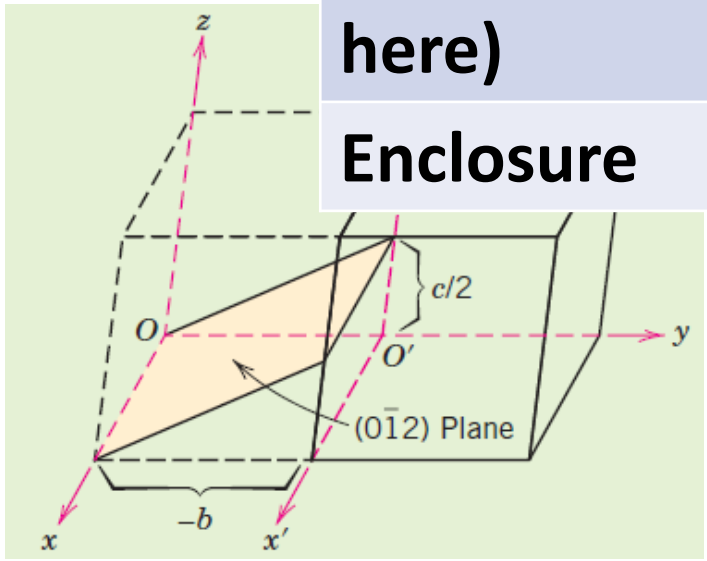
- 2 parallel planes are equivalent and have identical indices

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

- Determine the miller indices of t
  plane in the diagram shown bel





Summary			
	x	y	z
Intercepts	αa	- b	c/2
Intercepts in terms of lattice parameters (a, b, c)	α	-1	1/2
Reciprocals	1/α	-1	2
Reduction (not necessary here)			
Enclosure	$(0\ \bar{1}\ 2)$		

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

❖ *Hexagonal Crystals*

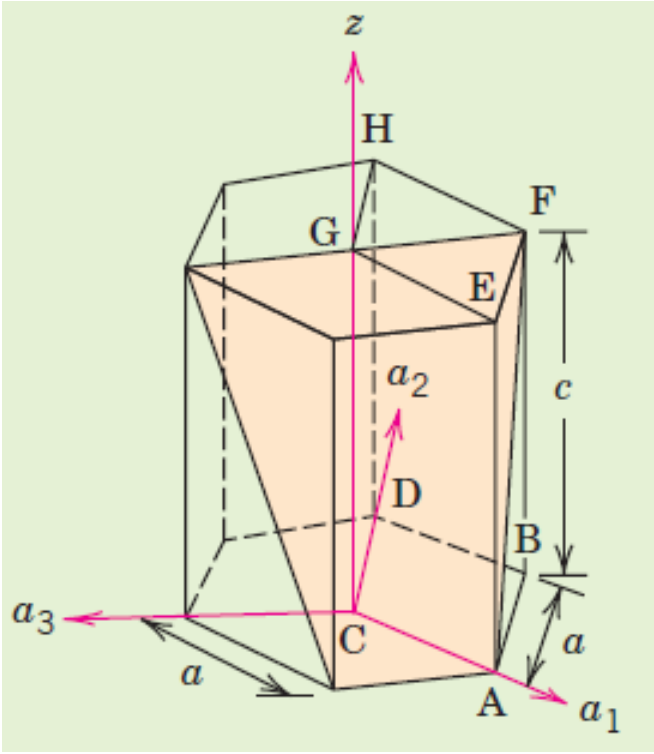
- A four-axis coordinate system otherwise called Miller-Bravais system is employed also
- Plane indices are denoted by 4 indices, as ***(h k i l)***
- ***i*** is some redundancy which is obtained by the sum of ***h*** and ***k*** through ***i = - (h + k)***
- Otherwise, *h*, *k* and *l* remains the same for both indexing systems i.e. the 3-axis (x, y, z) system or the Miller-Bravais 4-axis system

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Planes

❖ *Hexagonal Crystals*

- Determine the Miller-Bravais indices of the plane shown in the hexagonal unit cell below



THE STRUCTURE OF CRYSTALLINE SOLIDS

Planar Density – For Planes

- Planar density (PD) is taken as the number of atoms per unit area that are centered on a particular crystallographic plane, i.e.

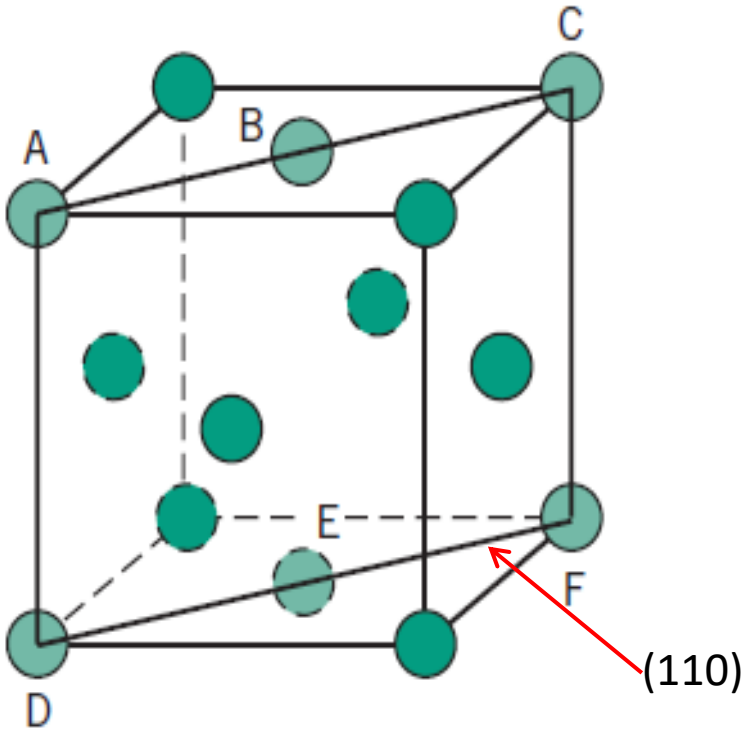
$$PD = \frac{number\ of\ atoms\ centered\ on\ plane}{area\ of\ plane}$$

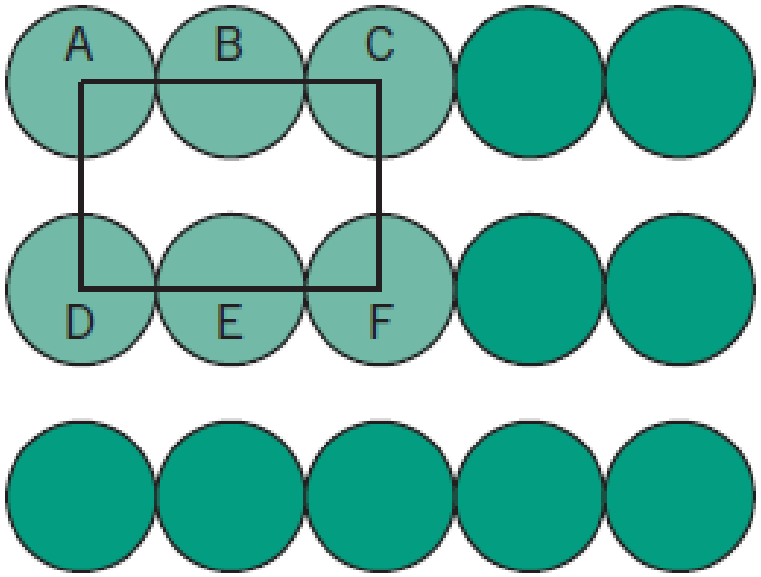
- Units reciprocal area (e.g. nm^{-2}, m^{-2})

THE STRUCTURE OF CRYSTALLINE SOLIDS

Planar Density

- Find the planar density of the (1 1 0) within an FCC unit cell





$$PD_{110} = \frac{2 \text{ atoms}}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}}$$

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Metallic Crystal Structures

❖ *Theoretical Density*

- A knowledge of crystal structure of metals makes it possible for its theoretical density to be calculated using the formula below:

$$\rho = \frac{\textbf{Mass of atoms in unit cell}}{\textbf{Total volume of unit cell}} = \frac{(\textbf{number of atoms/cell})(\textbf{atomic mass})}{(\textbf{volume of unit cell})(\textbf{Avogadro's constant})} = \frac{nA}{V_c N_A}$$

n is number of atoms, *A* and N_A are atomic weight and Avogadro's number respectively

Question

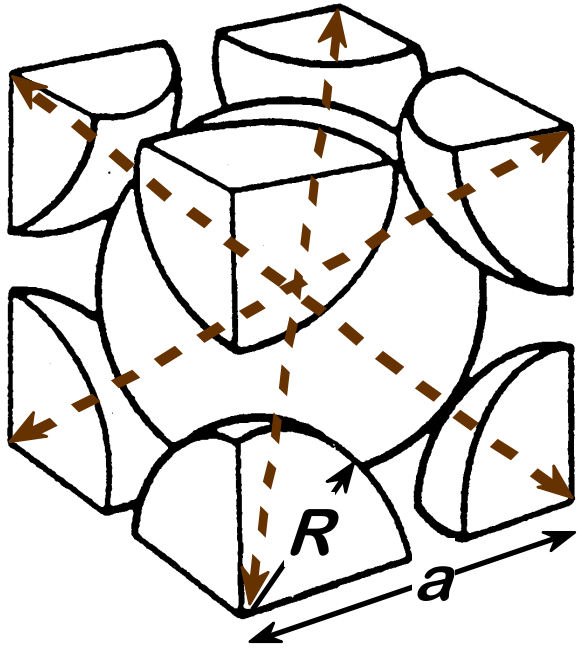
Cr has an atomic radius of 0.125 nm, a BCC crystal structure and atomic weight of 52.0 g/mol. Calculate its theoretical density (Measured density: 7.19 g/cm^3).

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Metallic Crystal Structures

❖ *Theoretical Density*



- Ex: Cr (BCC)

A = 52.00 g/mol

R = 0.125 nm

n = 2 atoms/unit cell

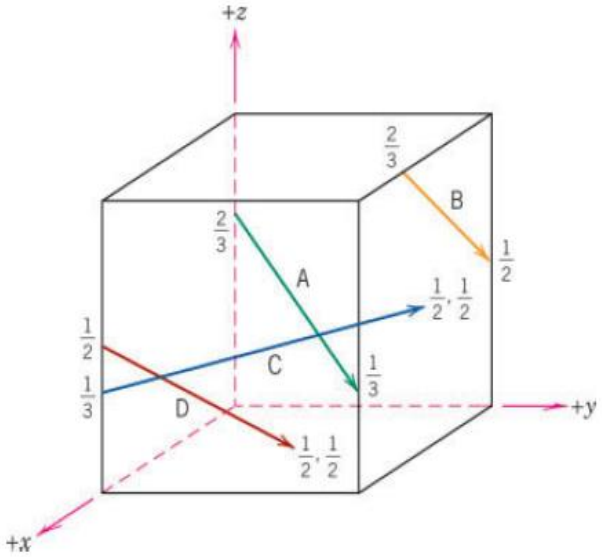
$a = 4R/\sqrt{3} = 0.2887$ nm

$\rho_{theoretical}$	= 7.18 g/cm^3
r_{actual}	= 7.19 g/cm^3

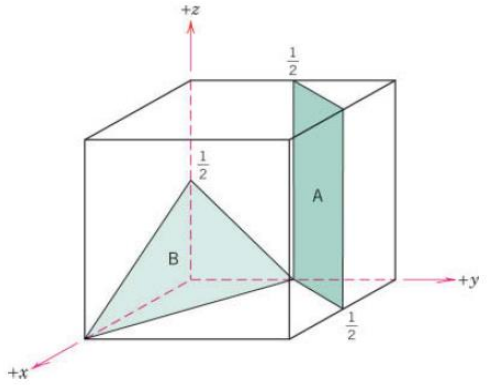
$$\rho = \frac{\frac{\text{atoms}}{\text{unit cell}} \times \frac{\text{g}}{\text{mol}}}{\frac{\text{volume}}{\text{unit cell}} \times \frac{\text{atoms}}{\text{mol}}} = \frac{2 \times 52.00}{a^3 \times 6.022 \times 10^{23}}$$

QUIZ

- Determine the indices for the directions shown in the following cubic unit cell



- Determine the Miller indices for the planes shown in the following unit cell



THE STRUCTURE OF CRYSTALLINE SOLIDS

Linear and Planar Densities

- Linear density (LD) and planar density (PD) are important in the process of *slip* which occurs in plastic deformation of metals
- Slip occurs on
  - most densely packed crystallographic planes and
  - in those planes, along the most dense crystallographic directions (having the greatest atomic packing)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Single Crystals

- In crystalline solids, when the periodic and repeated arrangement of atoms extends throughout the entirety of the specimen, we have a *single crystal*
- All the unit cells interlock and have the same orientation
- Single crystals exists naturally but can also be produced artificially
- When a single crystal is allowed to grow without constraint, it will assume a regular geometric shape having flat surfaces



Garnet single crystal

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Single Crystals

- Ordinarily difficult to grow because the environment must be carefully controlled

Polycrystalline Materials

- Most crystalline solids have a composition of so many small crystals also known as *grains*
- They are called *polycrystalline* materials

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Polycrystalline Materials

- Formation of grains arises during the solidification process of the polycrystalline specimen through the following processes:

  i. Very small grains also called *nuclei* with *random crystallographic orientations* develop at various sites in the initial stages

  ii. The small grains grow by successive *addition of atoms* from the surrounding liquid to the *structure* of each

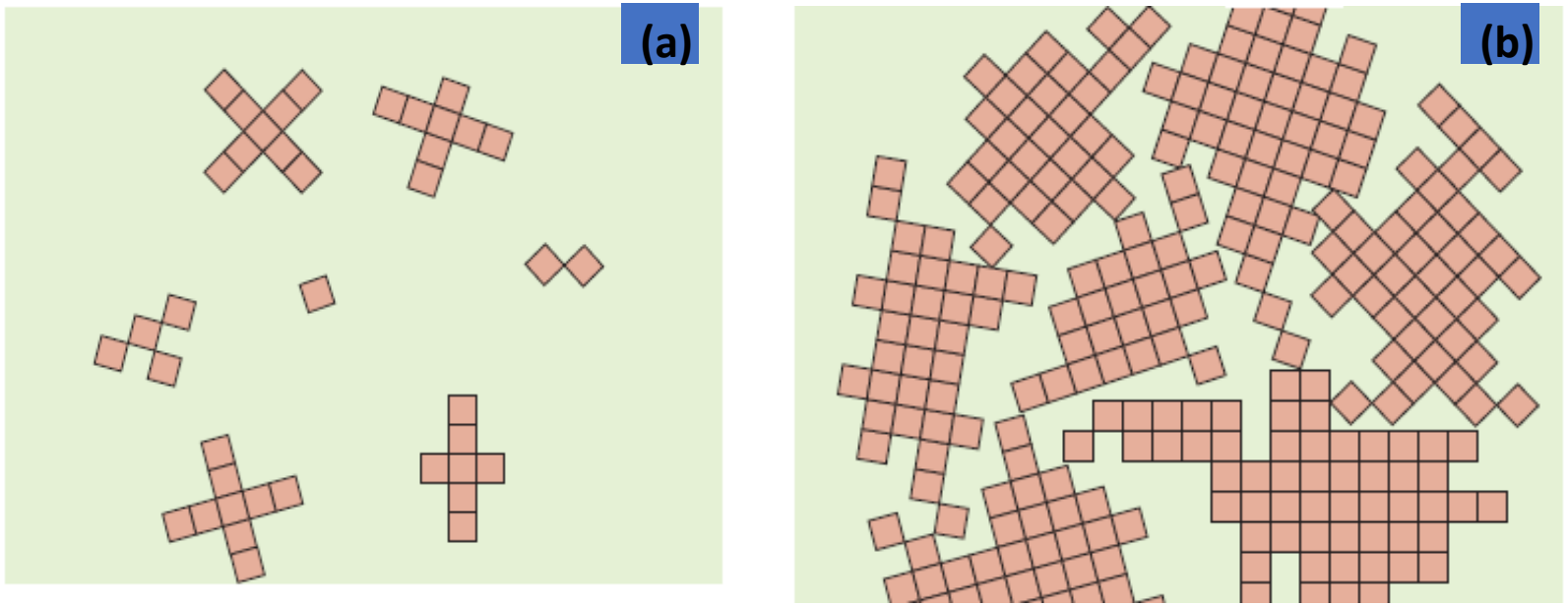
  iii. The extremities of adjacent grains impinge on each other as the solidification process nears completion

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Polycrystalline Materials

- Crystallographic orientation varies from grain to grain
- There is an atomic mismatch within the region where two grains meet. This region/area is called *grain boundary*

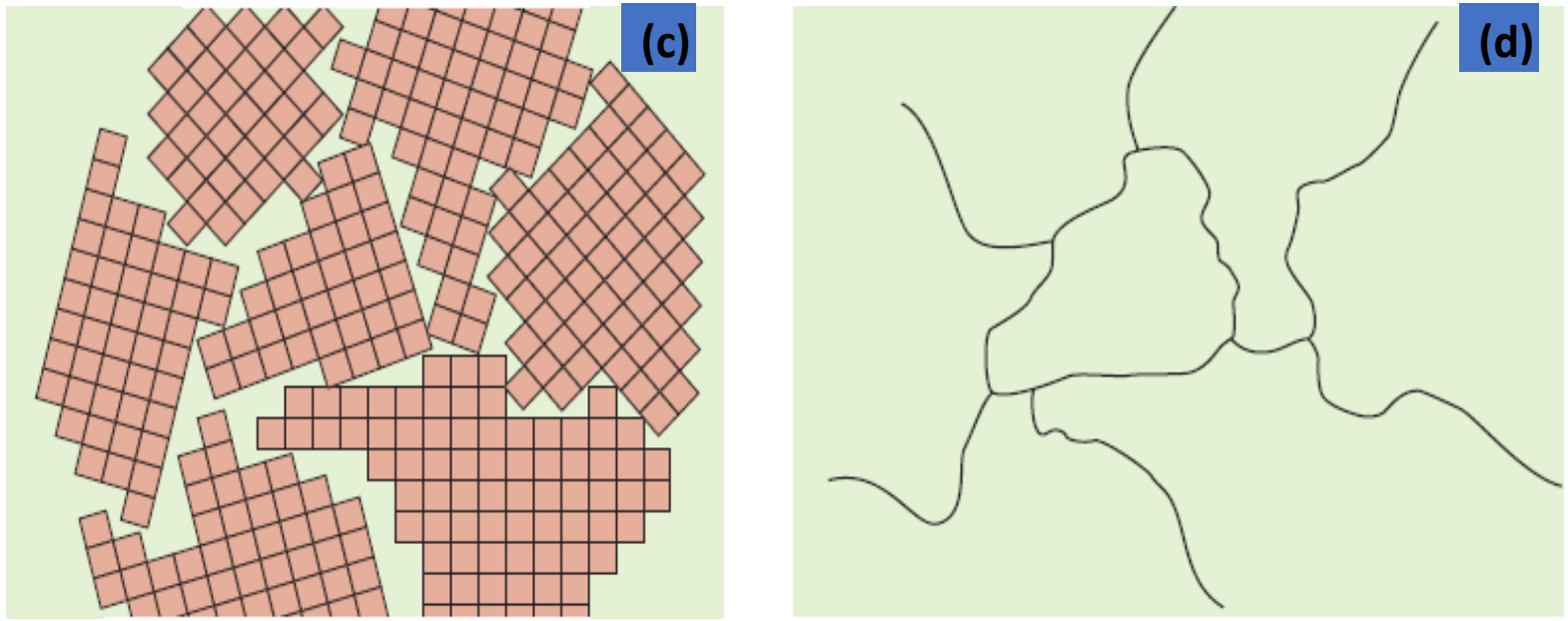


The schematic representation of the formation of a polycrystalline material is shown *(NB: Square grids depict unit cells)*

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Polycrystalline Materials



The schematic representation of the formation of a polycrystalline material is shown *(NB: Square grids depict unit cells)*



Photomicrograph of brass, 60×

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Anisotropy

- Some single crystals have some of their physical properties depending on the *crystallographic direction* along which the measurements were taken.
- A material property's dependence on the crystallographic direction along which it was measured is termed *anisotropy*
- It is associated with the *variance in atomic or ionic spacing* with crystallographic direction
- So, what is isotropy???

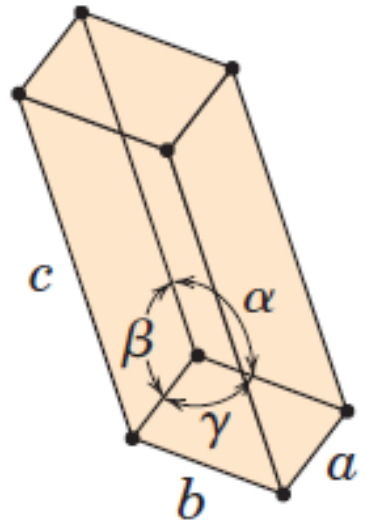
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Anisotropy

- The *degree* of anisotropy *increases* with *decrease* in *structural symmetry*
- Triclinic structured materials are highly anisotropic, why???

$a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma$



THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Anisotropy

- Polycrystalline materials - individual grains have random crystallographic orientation, hence anisotropic
- But the specimen composed of the grain aggregates behaves isotropically

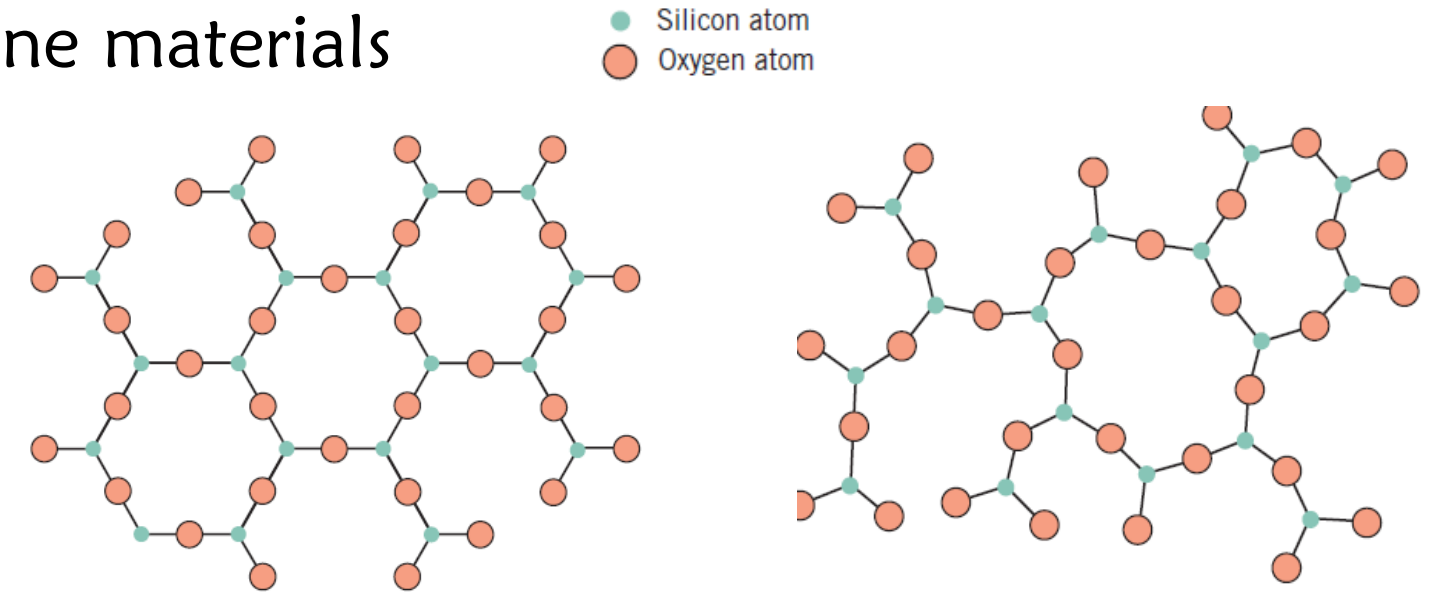
	Modulus of Elasticity (GPa)			Crystal Structure
Metal	[1 0 0]	[1 1 0]	[1 1 1]	
Aluminum	63.7	72.6	76.1	FCC
Copper	66.7	130.3	191.1	FCC
Iron	125.0	210.5	272.7	BCC
Tungsten	384.6	384.6	384.6	BCC

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Anisotropy

- The magnitude of the property measured represents some average of the directional values
- Texture: The crystallographic orientation preferential nature of polycrystalline materials



Crystalline silicon dioxide (left) and noncrystalline silicon dioxide (right)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

X-Ray Diffraction

- Information about the crystal structure of a material can be obtained by using the *x-ray diffraction* technique
- When a beam of x-rays with single wavelength on the same order of magnitude as the atomic spacing of a material strikes that material;

  i. X-rays are scattered in all directions
  ii. Most of the rays/radiations scattered by one atom are cancelled out by that of other atoms
  iii. However, x-rays that strike certain crystallographic planes at specific angles are reinforced rather than annihilated

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

X-Ray Diffraction

- The phenomenon of reinforcement of scattered rays is known as *diffraction*
- X-rays are diffracted (beam is reinforced) when *Bragg's law* is satisfied by conditions:

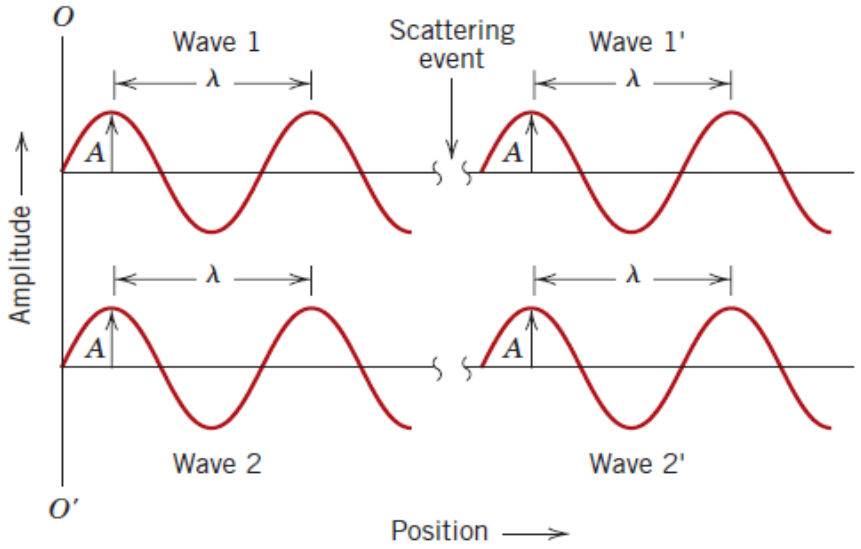
$$n\lambda = 2d_{hkl}\sin\theta$$

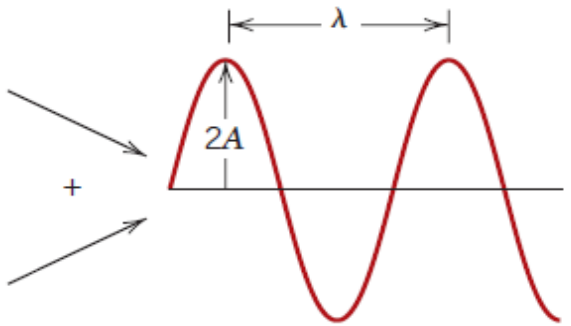
  - λ is the wavelength of the x-rays,
  - θ is half the angle between the diffracted beam and original beam
  - d_{hkl} is the interplanar spacing between the planes that cause constructive reinforcement of the beam
  - n is the order of reflection (1, 2, 3, 4, ……)

THE STRUCTURE OF CRYSTALLINE SOLIDS

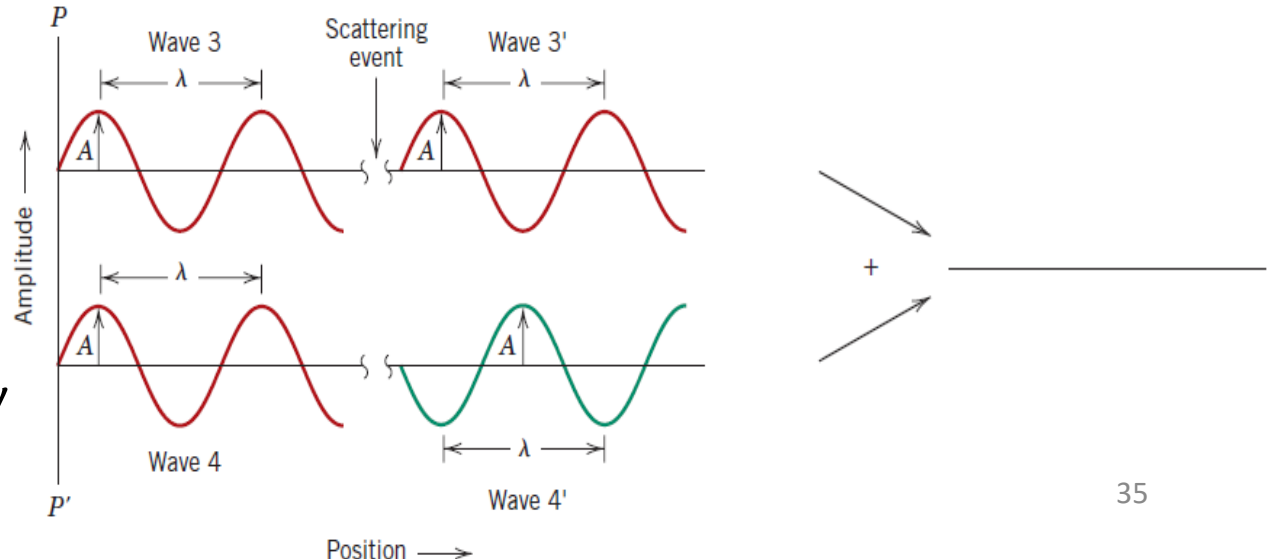
Crystalline Materials

X-Ra





Two waves (1 and 2) that have the same wavelength λ and remain in phase after a scattering event (waves 1′ and 2′) constructively interfere with one another. The amplitudes of the scattered waves add together in the resultant wave



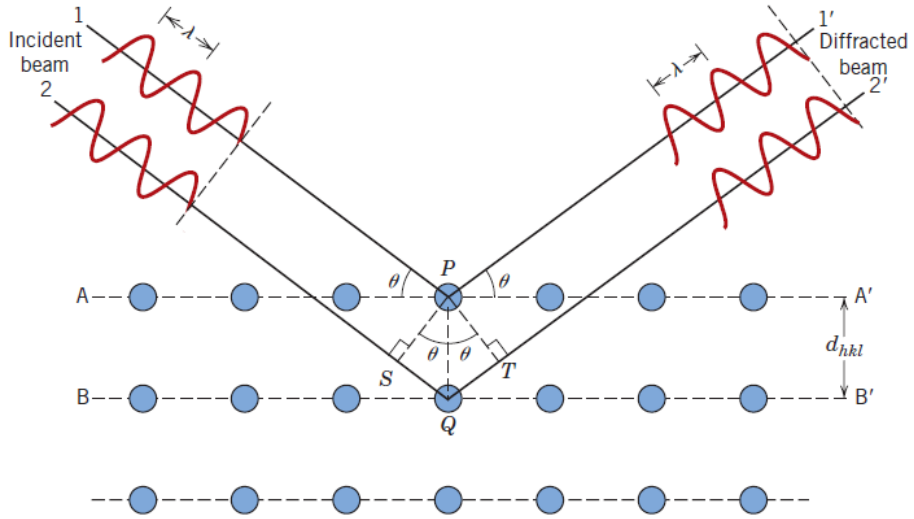
Two waves (3 and 4) that have the same wavelength λ and become out of phase after a scattering event (waves 3′ and 4′) destructively interfere with one another. The amplitudes of the two scattered waves cancel one another)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

X-Ray Diffraction

- Bragg's law: Relationship among λ, d and θ for constructive interference



Diffraction of x-rays by planes of atoms (A–A′ and B–B′)

$$n\lambda = \overline{SQ} + \overline{QT} \quad \Rightarrow \quad n\lambda = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2d_{hkl}\sin\theta$$

- The magnitude of the distance between two adjacent and parallel planes of atoms (i.e. d_{hkl}) is a function of the Miller Indices as well as the lattice parameter

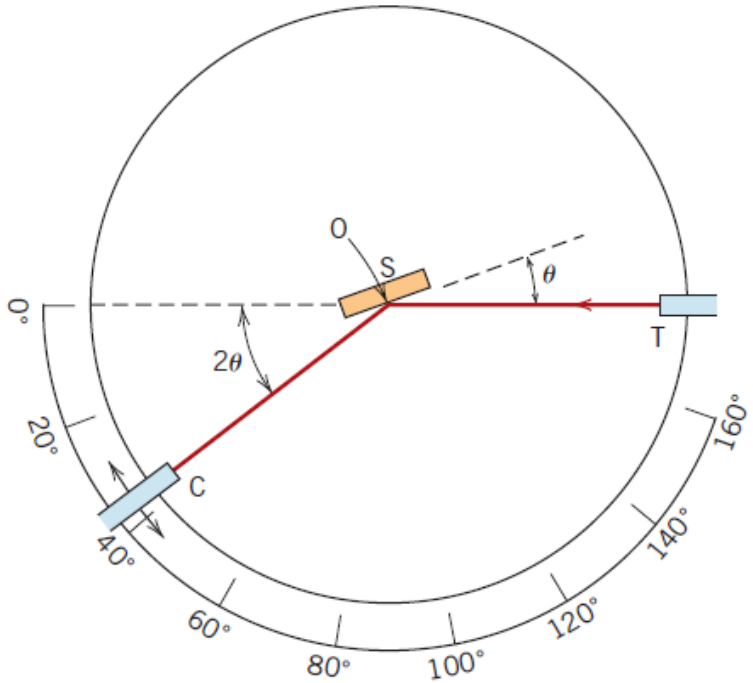
- For crystal structures with cubic symmetry, $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$

THE STRUCTURE OF CRYSTALLINE SOLIDS

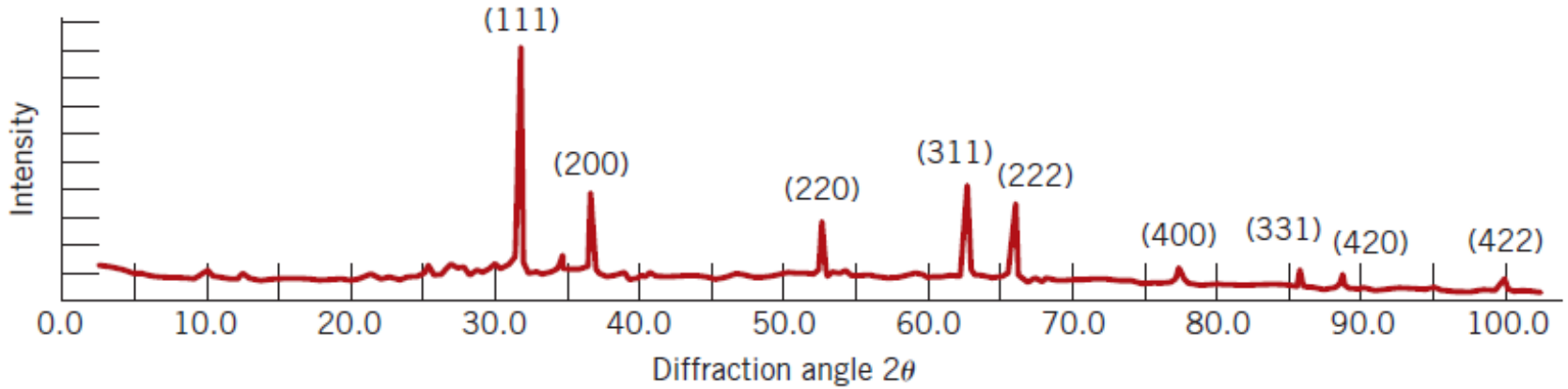
Crystalline Materials

X-Ray Diffraction

- Diffraction techniques – A common technique employs a powdered or polycrystalline specimen



Schematic diagram of an x-ray diffractometer; T x-ray source, S specimen, C detector, and O the axis around which the specimen and detector rotate.



Diffraction pattern for powdered lead (Pb)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

X-Ray Diffraction

- **Question**

For BCC Fe, its lattice parameter is 0.2866 nm. If a monochromatic radiation of wavelength 0.1790 nm is used and the order of reflection is 1, compute

a) the interplanar spacing, and

b) the diffraction angle for the (2 2 0) set of planes.

- **Solution**

a)

a = 0.2866 nm

h = 2, k = 2, and l = 0

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.2866\,nm}{\sqrt{(2)^2+(2)^2+(0)^2}} = 0.1013\,nm$$

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

X-Ray Diffraction

b) $$n\lambda = 2d_{hkl}\sin\theta \Rightarrow \sin\theta = \frac{n\lambda}{2d_{hkl}} = \frac{(1)(0.1790\,nm)}{(2)(0.1013\,nm)} = 0.8840$$

$$\theta = \sin^{-1}(0.8840) = 62.13°$$

The diffraction angle is 2θ or 2θ = (2)(62.13°) = 124.26°

Crystal Structure Identification

- From the diffraction patterns recorded, the orientation and quality of the single crystal can be determined.
- The crystal structure can also be determined using the rotating crystal and fixed wavelength x-ray source.

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Crystal Structure Identification

- In SC metals, all possible planes will diffract, giving an $h^2 + k^2 + l^2$ pattern of 1, 2, 3, 4, 5, 6, 8, ……
- In BCC metals, diffraction occurs only from planes having an even $h^2 + k^2 + l^2$ sum of 2, 4, 6, 8, 10, 12, 14, 16, ……
- For FCC metals, more destructive interference occurs, and planes having $h^2 + k^2 + l^2$ sums of 3, 4, 8, 11, 12, 16, …… will diffract.
- By calculating the values of sin ϑ and then finding the appropriate pattern, the crystal structure can be determined for metals having one of these simple structures

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Crystal Structure Identification

- **Example**

The results of a x-ray diffraction experiment using x-rays with $\lambda = 0.7107$ Å (a radiation obtained from a molybdenum (Mo) target) show that diffracted peaks occur at the following 2θ angles:

Peak	2θ	Peak	2θ
1	20.20	5	46.19
2	28.72	6	50.90
3	35.36	7	55.28
4	41.07	8	59.42

Determine the crystal structure, the indices of the plane producing each peak, and the lattice parameter of the material.

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Crystal Structure Identification

- We can first determine the $\sin^2\theta$ value for each peak, then divide through by the lowest denominator, 0.0308.

- Use 2θ values for any of the peaks to calculate the inter-planar spacing and thus the lattice parameter. Picking peak 8:

$$d_{400} = \frac{\lambda}{2\sin\theta} = \frac{0.7107}{2\sin 29.71} = 0.7170\ \text{Å}$$

$$a_o = d_{400}\sqrt{h^2+0+0} = (7170\ \text{Å})(4) = 2.868\ \text{Å}$$

Peak	2θ	$\sin^2\theta$	$\frac{\sin^2\theta}{0.0308}$	$h^2+k^2+l^2$	(hkl)
1	20.20	0.0308	1	2	(110)
2	28.72	0.0615	2	4	(200)
3	35.36	0.0922	3	6	(211)
4	41.07	0.1230	4	8	(220)
5	46.19	0.1539	5	10	(310)
6	50.90	0.1847	6	12	(222)
7	55.28	0.2 152	7	14	(321)
8	59.42	0.2456	8	16	(400)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Crystal Structure Identification

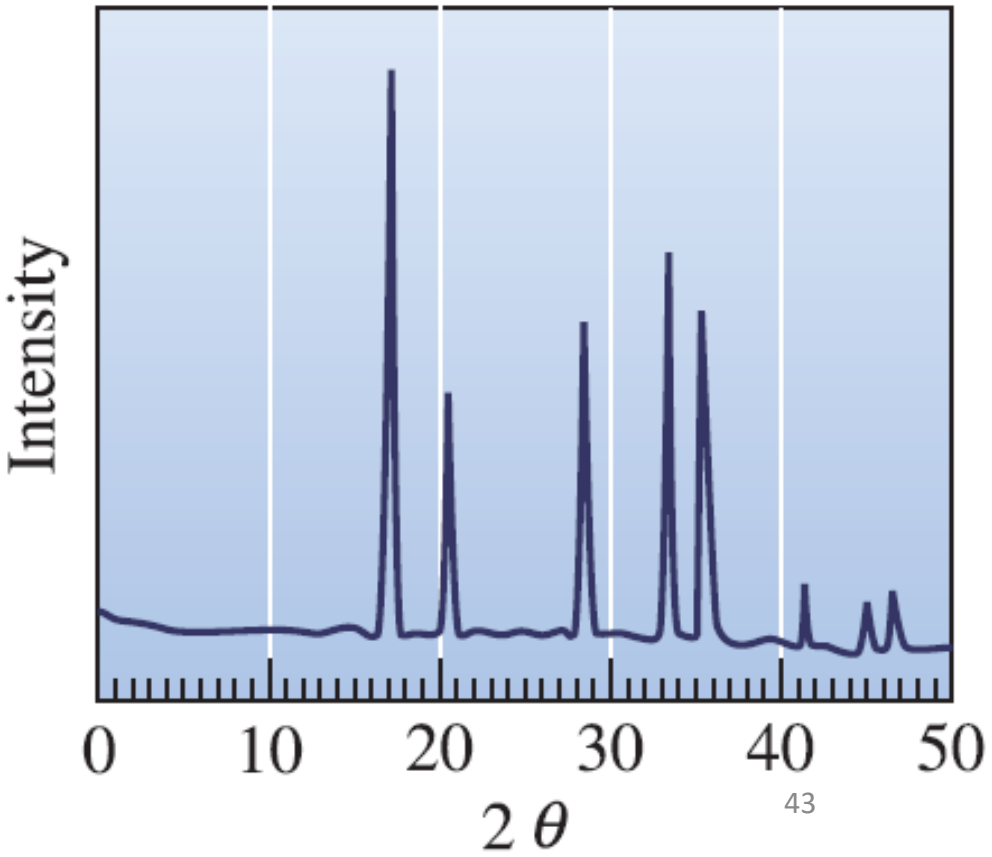
- **Example**

The figure shows the results of an x-ray diffraction experiment in the form of the intensity of the diffracted peak values, the 2θ diffraction angle. If x-rays with a wavelength of 0.15418 nm are used, determine

a) the crystal structure of the metal;

b) the indices of the planes that produce each of the peaks; and

c) the lattice parameter of the metal.



THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystalline Materials

Crystal Structure Identification

Peak	2θ	$\sin^2\theta$	$\frac{\sin^2\theta}{0.023}$	$h^2+k^2+l^2$	(hkl)	d_{hkl}	$A_{o(hkl)}$
1	17.5	0.023	1.000	3	(111)	0.507	0.878
2	20.5	0.032	1.368	4	(200)	0.433	0.867
3	28.5	0.061	2.618	8	(220)	0.313	0.886
4	33.5	0.083	3.589	11	(311)	0.268	0.887
5	35.5	0.093	4.016	12	(222)	0.253	0.876
6	41.0	0.123	5.300	16	(400)	0.220	0.880
7	45.0	0.146	6.328	19	(331)	0.201	0.878
8	46.5	0.156	6.734	20	(420)	0.196	0.873

a) The 3,4,8,11, . . . sequence means that the material is FCC

b) $a_o = \frac{0.878+0.867+0.886+0.887+0.876+0.880+0.878+0.873}{8} = 0.878$